Foundations of Deep Learning **Fall 2026**

Exercise 4: Optimization Landscape of Neural Networks

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Problem 1 (Network near initialization):
Given an input vector $\mathbf{x} \in \mathbb{R}^d$, consider a shallow neural network defined by

$$f(\mathbf{W}) := \sum_{j=1}^{m} a_j \sigma(\mathbf{w}_j^\top \mathbf{x}), \qquad \mathbf{W} := \begin{bmatrix} \leftarrow \mathbf{w}_1^\top \rightarrow \\ \vdots \\ \leftarrow \mathbf{w}_m^\top \rightarrow \end{bmatrix} \in \mathbb{R}^{m \times d},$$

where σ is an activation function, $\mathbf{a} \in \mathbb{R}^m$ is a fixed (not trainable) vector of weights initialized such that $a_j \overset{\text{i.i.d.}}{\sim} \text{Unif}\left(\pm \frac{1}{\sqrt{m}}\right)$ [1], and $\mathbf{W} \in \mathbb{R}^{m \times d}$ are trainable weights.

We will consider the linearization of the function f around some initial weights $\mathbf{W}_0$ defined by

$$f_0(\mathbf{W}) = f(\mathbf{W}_0) + \langle \nabla f(\mathbf{W}_0), \mathbf{W} - \mathbf{W}_0 \rangle_F \,, \tag{1}$$

where $\langle \mathbf{A}, \mathbf{B} \rangle_F = \text{tr}(\mathbf{A}^\top \mathbf{B})$ is the Frobenius inner product.

a) Show that

$$\nabla f(\mathbf{W}) = \mathbf{D}\mathbf{a}\mathbf{x}^\top, \text{ where } \mathbf{D} = \text{diag}(\sigma'(\mathbf{w}_i^\top \mathbf{x})) = \begin{pmatrix} \sigma'(\mathbf{w}_1^\top \mathbf{x}) & & & \\ & \sigma'(\mathbf{w}_2^\top \mathbf{x}) & & \\ & & \cdots & \\ & & & \sigma'(\mathbf{w}_m^\top \mathbf{x}) \end{pmatrix}.$$

b) Show that

$$\langle \nabla f(\mathbf{W}_0), \mathbf{W} - \mathbf{W}_0 \rangle_F = \sum_{j=1}^{m} a_j \sigma'(\mathbf{w}_{0,j}^\top \mathbf{x})(\mathbf{w}_j - \mathbf{w}_{0,j})^\top \mathbf{x},$$

where $\mathbf{W}_0 = \begin{bmatrix} \leftarrow \mathbf{w}_{0,1}^\top \rightarrow \\ \vdots \\ \leftarrow \mathbf{w}_{0,m}^\top \rightarrow \end{bmatrix}$ denotes the weight at initialization.

c) Now assume the activation function σ is β-smooth, which satisfies:

$$|\sigma(r) - \sigma(s) - \sigma'(s)(r - s)| \leq \frac{\beta (r - s)^2}{2}$$

for all $r, s \in \mathbb{R}$. Show that for any $\mathbf{W} \in \mathbb{R}^{m \times d}$,

$$|f(\mathbf{W}) - f_0(\mathbf{W})| \leq \frac{\beta}{2\sqrt{m}} \|\mathbf{W} - \mathbf{W}_0\|_F^2 \, \|\mathbf{x}\|^2.$$

d) What do you conclude about the role of over-parametrization?

[1] Equivalently, one can define $f(\mathbf{W}) := \frac{1}{\sqrt{m}} \sum_{j=1}^{m} a_j \sigma(\mathbf{w}_j^\top \mathbf{x})$ with $a_j \overset{\text{i.i.d.}}{\sim} \text{Unif}(\pm 1)$.